Table S4 Analysis populations and score availability

A. Directional and chemistry populations

Target pair	Direction	Labeled n	S1 directional n	S1 missing n	Unassigned fold after S1	Chemistry OOF n
EGFR/HER2	Dual vs A-only	65	65	0	0	65
EGFR/HER2	Dual vs B-only	60	60	0	1	59
JAK1/JAK2	Dual vs A-only	64	64	0	0	64
JAK1/JAK2	Dual vs B-only	64	64	0	0	64
JAK1/TYK2	Dual vs A-only	64	64	0	1	63
JAK1/TYK2	Dual vs B-only	64	64	0	1	63
PIK3CA/mTOR	Dual vs A-only	64	64	0	0	64
PIK3CA/mTOR	Dual vs B-only	64	64	0	0	64
AChE/BChE	Dual vs A-only	55	52	3	0	52
AChE/BChE	Dual vs B-only	56	55	1	0	55
F2/F10	Dual vs A-only	64	64	0	1	63
F2/F10	Dual vs B-only	64	64	0	1	63
PPARG/PPARA	Dual vs A-only	64	64	0	1	63
PPARG/PPARA	Dual vs B-only	64	64	0	0	64
PPARA/PPARD	Dual vs A-only	64	64	0	0	64
PPARA/PPARD	Dual vs B-only	64	64	0	0	64

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S10B.

Table S4 Analysis populations and score availability

B. Score availability across activity classes

Target pair	Scheme	Expected A	Valid A	Missing A	Expected B	Valid B	Missing B
EGFR/HER2	S1	109	109	0	109	109	0
EGFR/HER2	S2	109	109	0	109	109	0
EGFR/HER2	S3	109	109	0	109	109	0
EGFR/HER2	S4	109	109	0	109	109	0
EGFR/HER2	S5	109	109	0	109	109	0
JAK1/JAK2	S1	110	110	0	110	110	0
JAK1/JAK2	S2	110	110	0	110	110	0
JAK1/JAK2	S3	110	110	0	110	110	0
JAK1/JAK2	S4	110	110	0	110	110	0
JAK1/JAK2	S5	110	110	0	110	110	0
JAK1/TYK2	S1	110	110	0	110	110	0
JAK1/TYK2	S2	110	110	0	110	110	0
JAK1/TYK2	S3	110	110	0	110	110	0
JAK1/TYK2	S4	110	110	0	110	110	0
JAK1/TYK2	S5	110	110	0	110	110	0
PIK3CA/mTOR	S1	110	110	0	110	110	0
PIK3CA/mTOR	S2	110	110	0	110	110	0
PIK3CA/mTOR	S3	110	110	0	110	110	0
PIK3CA/mTOR	S4	110	110	0	110	110	0
PIK3CA/mTOR	S5	110	110	0	110	110	0
AChE/BChE	S1	98	95	3	98	95	3
AChE/BChE	S2	98	94	4	98	94	4
AChE/BChE	S3	98	95	3	98	95	3
AChE/BChE	S4	98	94	4	98	94	4
AChE/BChE	S5	98	97	1	98	97	1
F2/F10	S1	110	110	0	110	110	0
F2/F10	S2	110	109	1	110	110	0
F2/F10	S3	110	110	0	110	110	0
F2/F10	S4	110	109	1	110	110	0
F2/F10	S5	110	108	2	110	108	2

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S10B.

Table S4 Analysis populations and score availability (continued)

B. Score availability across activity classes

Target pair	Scheme	Expected A	Valid A	Missing A	Expected B	Valid B	Missing B
PPARG/PPARA	S1	110	110	0	110	110	0
PPARG/PPARA	S2	110	108	2	110	108	2
PPARG/PPARA	S3	110	110	0	110	110	0
PPARG/PPARA	S4	110	108	2	110	108	2
PPARG/PPARA	S5	110	110	0	110	110	0
PPARA/PPARD	S1	110	110	0	110	110	0
PPARA/PPARD	S2	110	105	5	110	105	5
PPARA/PPARD	S3	110	110	0	110	110	0
PPARA/PPARD	S4	110	105	5	110	105	5
PPARA/PPARD	S5	110	110	0	110	110	0

Panel A counts ligand-direction records. Panel B aggregates the four activity classes; class-specific missingness, exclusions and ligand identifiers remain in the complete source files. Dual vs A-only means dual versus A-only using target B; Dual vs B-only uses target A. Missing fold assignments are excluded from chemistry evaluation. Paired method comparisons use their own complete-case populations, reported in Table S10B.